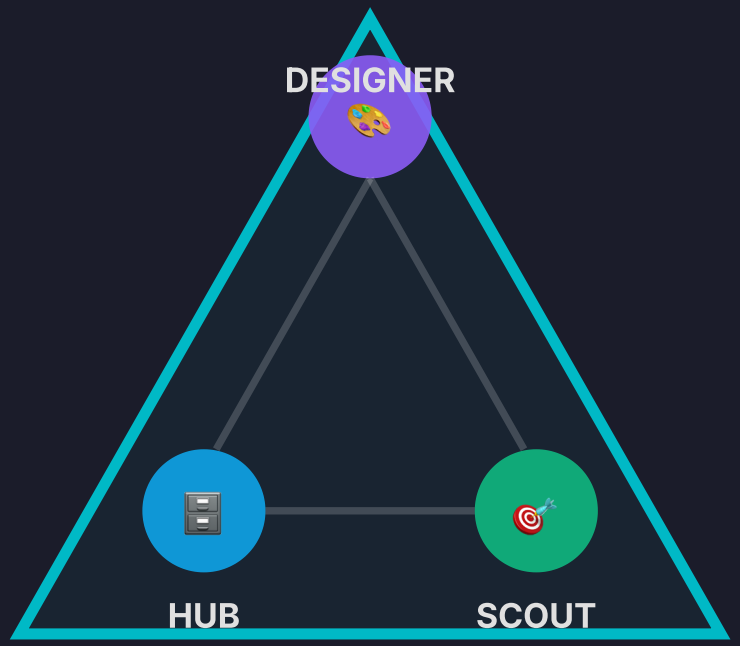
Team Details & Artefact links



Triangle Smart Targeting
& Real-Time Activation

TriStar

EPAM HACKATHON 2026 · LOYALTY AI PLATFORM

TEAM

Representative: *Ram Naresh, Kartikey Puri*

TEAM MEMBERS

RN	**Ram Naresh** — AI Architecture & Backend	Rep
KP	**Kartikey Puri** — Full Stack & Integration	Rep
MS	**Mohit Soni** — Frontend & UX Design	
RS	**Riyaj Shaikh** — Backend & DevOps	

🎨 **Designer — AI Offer Copilot**

🗄️ **Hub — Lifecycle State**

🎯 **Scout — Real-Time Activation**

What we built: AI-powered three-layer loyalty engine — **Designer** turns plain-English objectives into fraud-validated OfferBriefs; **Hub** tracks lifecycle in Redis; **Scout** activates at the perfect moment using GPS, weather & behaviour signals. Stack: **React 19 · FastAPI · Claude API · Redis · Azure**.

git.garage.epam.com/kartikey_puri/tristar

EPAM Garage · develop branch · Full source code

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Problem Statement

The current loyalty ecosystem operates reactively — members accumulate points but receive limited, generic engagement:

📉	⏱️	🎯
73% offers go unnoticed	**3–5h** manual offer creation	**0%** real-time context used

- High-value lapsed members remain completely unengaged
- Offers ignore location, weather & purchase behaviour
- No feedback loop → repeated campaign inefficiencies
- Fraud & over-discounting go undetected before launch
- Siloed tools: no lifecycle visibility draft-to-expiry

😤 Generic Offers → 🙈 Member Ignores → 💸 Revenue Lost

CATEGORY

AI-Powered Loyalty Personalisation

Aligned with Customer Experience & Digital Innovation

<epam>

Solution

TriStar transforms loyalty into an **AI-powered, real-time engagement engine** — delivering the **right offer** to the **right member** at the **right moment**.

SYSTEM OVERVIEW — END-TO-END FLOW

🎨 LAYER 1 · DESIGNER

Biz Objective / Inventory Hint / Live Deal Feed → 🤖 Claude Sonnet 4.6 → OfferBrief JSON → ◇ Fraud Check

✓ Pass → Hub (draft) | ✗ Critical → Block

↓ approved

🗄️ THE HUB — SHARED CONTEXT STATE

Redis / In-Memory · draft → approved → active → expired · Audit Log · Expiry Sweep 300s

↓ context match

🎯 LAYER 3 · SCOUT — REAL-TIME ACTIVATION ENGINE

Flow A	Flow B	Flow C	Flow D
Purchase Event	**Hub Match**	**Partner Trigger**	**Self-Activation**
HMAC webhook · Determ. ≥ 70	Claude AI > 60 · CASL dedup	Claude Haiku · 24h brief	Customer-accept JWT

↓

📱 Push Notification · ⏰ Queue for Later (8am) · 📋 Activation Log

Tech Stack

Next.js 15 | FastAPI | Claude Sonnet 4.6 | Redis | Azure | Pydantic v2

Impact

+25%	**80%**	**<200ms**
lapsed member reactivation	faster campaign creation	real-time activation

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Benefits

80%	+25%	100%
Faster Campaigns	Member Reactivation	Audit Ready

For Marketers

- Generate validated OfferBriefs in seconds — not hours
- Built-in fraud detection catches over-discounting before launch
- Inventory-driven recommendations increase campaign ROI
- Real-time risk flags surfaced before any offer goes live

For Members

- Offers arrive only when GPS, time & weather align
- Fewer, more relevant notifications — less fatigue
- Seamless, personalised loyalty experience at scale

For the Business

- Full lifecycle visibility: draft → approved → active → expired
- Immutable audit trail — OWASP & compliance ready
- Horizontal scale on Azure · millions of members

Future Enhancements

PRODUCT ROADMAP

Phase 1 MVP	Phase 2 Intelligence	Phase 3 Scale & Reach
Now	Q3 2026	Q1 2027

Phase 1 — MVP ✓

Now · Hackathon

- Designer + Hub + Scout core
- Claude API brief generation
- 4-pattern fraud detection
- Real-time context scoring
- JWT auth + Redis + Azure

Phase 2 — Intelligence

Q3 2026

- ML-based segment targeting
- A/B testing framework
- Bilingual AI-generated offers
- Live analytics dashboard

Phase 3 — Scale & Reach

Q1 2027

- Partner ecosystem open APIs
- Event-driven low-latency arch
- Mobile SDK + geofencing
- Multi-agent autonomous campaigns

Live Now
- ✓ 3-layer AI engine
- ✓ Fraud detection
- ✓ Real-time scoring
- ✓ Redis + Azure

Next Up
- ✓ ML targeting
- ✓ A/B testing
- ✓ Bilingual offers
- ✓ Analytics dash

Future
- ✓ Partner APIs
- ✓ Mobile SDK
- ✓ Geofencing
- ✓ Multi-agent AI

Built-in Guardrails (Phase 1)

- 1 offer / member / hr
- No duplicates / 24h
- Quiet hrs 10 pm–8 am
- Critical fraud → hard block

Security: OWASP Top 10 mitigations built in — parameterised queries, Zod + Pydantic input validation, JWT 1h expiry, Azure Key Vault for secrets, PII-free logs (member_id only).

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